Mehmet Efe Lorasdagi Curriculum Vitae

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GitHub Google Scholar

Education

UCLA Sep 2025 – Present
Ph.D. in Electrical and Computer Engineering GPA: 4.0/ 4.0 *Los Angeles, California*
Advisor: Prof. Vwani Roychowdhury
Research Interests: Large Language Model Post-Training, Reinforcement Learning, Neural Signal Processing, Continual Learning

Bilkent University Ankara, Turkey
M.S. in Electrical and Electronics Engineering; GPA: 3.96/4.0 *Sep 2023 – Jul 2025*
Advisor: Prof. Suleyman S. Kozat
B.S. in Electrical and Electronics Engineering; GPA: 3.93/4.0 *Sep 2019 – Jul 2023*

Journal Papers

1. **M. E. Lorasdagi**, Dogan Can Cicek, Furkan Burak Mutlu, and S. S. Kozat, "Enhancing Deep Deterministic Policy Gradients on Continuous Control Tasks with Decoupled Prioritized Experience Replay". Under review.
2. **M. E. Lorasdagi**, A. B. Koc, A. T. Koc and S. S. Kozat, "Fitting Multiple Machine Learning Models With Performance Based Clustering". *IEEE Signal Processing Letters* (2025): 10.1109/LSP.2025.3537997.
3. **M. E. Lorasdagi**, M. Y. Turali, and S. S. Kozat. "Binary feature mask optimization for feature selection." *Neural Computing & Applications* (2024): https://doi.org/10.1007/s00521-024-10913-9
4. M. Y. Turali, **M. E. Lorasdagi**, and S. S. Kozat. "AFS-BM: enhancing model performance through adaptive feature selection with binary masking." *Signal, Image and Video Processing* (2024): 1-12.

Conference Papers

1. Lina Zhang, Tonmoy Monsoor, **M. E. Lorasdagi**, Prateik Sinha, Chong Han, Peizheng Li, Yuan Wang, Jessica Pasqua, Colin McCrimmon, Rajarshi Mazumder, and Vwani Roychowdhury, "Can Multimodal Large Language Models Understand Pathologic Movements? A Pilot Study on Seizure Semiology." *48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2026. Accepted for poster presentation. https://arxiv.org/abs/2605.03352
2. Mehmet Yigit Turali, Shreyas Rajesh, **Mehmet Efe Lorasdagi**, Chenda Duan, Kartik Sharma, and Vwani Roychowdhury, "PANINI-DR: Controlled Deep Research Through Generative Semantic Workspaces." *Under review, ICLR 2027*. https://openreview.net/forum?id=isVALiRJQL
3. Tonmoy Monsoor, Colin M. McCrimmon, Prateik Sinha, Lina Zhang, Manoj Reddy Dareddy, **Mehmet Efe Lorasdagi**, Ranmal A. Samarasinghe, and Vwani Roychowdhury, "Self-Exciting Population Event Models Reveal Abnormal Temporal Amplification in MAPT-Mutant Human Brain Assembloids." *Submitted to ICASSP 2027*. bioRxiv preprint, 2026. https://doi.org/10.64898/2026.10.01.756068

Work Experience

Machine Learning Engineer Sep 2023 – Aug 2025
Databoss

- Worked on the development of spatiotemporal prediction frameworks for crime and traffic accident prediction for gendarmerie
- Created a pipeline for processing data and training an ensemble of models on the data, including LightGBM, Linear Regression, and Transformers
- Worked on the development of agentic RAG systems for various applications using LangChain framework
- Developed information retrieval systems, using an ensemble approach where documents are retrieved based on topic, subtopic and text embedding similarity
- Organized and managed labelers for creating the datasets for the information retrieval system
- Worked on developing agentic LLM systems that perform information retrieval, generate answers and reduce hallucinations in generated answers, inspired by Self-RAG

Research Intern Jun 2022 – Aug 2022
Fraunhofer IIS *Erlangen, Germany*

- Analyzed the text-to-speech system developed at Fraunhofer IIS with StyleMelGAN vocoder architecture
- Modified the system to be compatible with the Apache TVM framework by redefining convolution and random number generation operations in the code
- Optimized the modified StyleMelGAN vocoder using Apache TVM framework on NVIDIA GTX 1660 GPU by decreasing the inference time by half

Academic Duties

Academic Mentor Jun 2026 – Aug 2026
Aerospace Corporation project at RIPS program *IPAM*

Reviewer 2024 – Present
Springer Neural Computing and Applications

Teaching Assistant Fall 2023 – Spring 2025
EEE493-494 Industrial Design Project *Bilkent University*

Related Coursework

Convex Optimization, Information Theory, Neural Signal Processing, Linear Programming, Natural Language Processing, Nonlinear Control, Deep Generative Networks, Detection and Estimation Theory, Reinforcement Learning and Dynamic Programming, Random Processes, Digital Signal Processing, Neural Networks, Telecommunications, Statistical Learning and Data Analytics, Introduction to Machine Learning, Feedback Control Systems

Selected Research Projects

Deep Learning-Based Spike Sorting Ongoing
UCLA

- Developing deep learning methods for spike sorting, targeting faster processing and improved accuracy.

Synthesizing Realistic Images from Sketches with GANs Spring 2024

- Explored synthesizing realistic images from sketches using Generative Adversarial Networks (GANs), focusing on Conditional GANs (cGANs) and StyleGAN
- By conditioning the generator and discriminator on edge maps derived from sketches, we examined the effectiveness of various conditioning approaches, including training on Canny and HED edge maps
- Our experiments showed that, while conditioning improved certain contextual aspects, achieving photorealism remained challenging

Continually Learning Video Labeling System Fall 2022 – Spring 2023

- Developed a system that facilitates the user for labeling video to create annotated custom datasets
- Evaluated the performance and implemented YOLOv7 object detection algorithm combined with SORT algorithm for tracking
- Evaluated and implemented iCaRL and GDumb methods for catastrophic forgetting prevention in online learning
- Implemented online learning strategies that enable continual training of the system under a stream of user-corrected photos
- Achieved increased labeling speed in the system by online learning strategies

Awards & Honors

Departmental Award Fall 2025 – Present
UCLA ECE Department *PhD Fellowship*

5G and Beyond Graduate Fellowship Oct 2024 – Aug 2025
Turk Telekom & Information and Communication Technologies Authority

Comprehensive Scholarship 2023 – Jul 2025
Bilkent University *Merit-based tuition waiver and stipend during MS*

Academic Excellence Award Jun 2024
Bilkent University

Ranked 22nd among 150k university graduates in ALES (National GRE) Apr 2023
Council of Higher Education of Turkey

High Honor Student Jan 2020 – Jun 2023
Bilkent University *High Honor Student in all semesters during undergraduate studies*

Comprehensive Scholarship 2019 – 2023
Bilkent University *Merit-based tuition waiver and stipend during BS*

Ranked 42nd out of 2.5 million in University Entrance Exam 2019
Council of Higher Education of Turkey

Ranked first place in High School Physics Olympiad among 100 student groups 2019
Atilim University

Specialized Skills

Programming: Python, MATLAB
Machine Learning: PyTorch, TensorFlow, scikit-learn
Scientific Computing: NumPy, pandas
Tools: Linux, Git, Docker, LaTeX
Languages: Turkish (native), English
Test Scores: TOEFL iBT 112/120, GRE 159/170/4.5